MODULE 1 · CPDLC PRINCIPLES

# Lesson 02: Data Link Services

Lesson 2 of 6 · about 6 min

## LEARNING OBJECTIVES

- Describe the data link services available to the flight crew.
- Explain the logon (DLIC) process.
- Identify the services mandated in Europe.



Log on before entering CPDLC airspace.

## The services

| Service | Purpose |
|---|---|
| **DLIC** — Data Link Initiation Capability | Provides the information needed to establish data link communications between an ATS unit and the aircraft. It is done **before any other data link service** (the logon). |
| **ACM** — ATC Communications Management | Automated assistance for the transfer of ATC communications (voice and CPDLC) between sectors and units. |
| **ACL** — ATC Clearances | Operational exchanges: pilots send requests and reports; controllers issue clearances, instructions and notifications. |
| **AMC** — ATC Microphone Check | Lets the controller send one instruction to all CPDLC aircraft on a frequency to check that none is blocking the voice channel (stuck microphone). |
| **DCL** — Departure Clearance | Requesting and delivering departure clearances by data link. |
| **DSC** — Downstream Clearance | Obtaining a clearance from an ATS unit that is not yet in control of the aircraft, when it cannot be obtained through unit-to-unit coordination. |

The European data link rules mandated the first four services: **DLIC, ACM, ACL and AMC**.

## Logon

- The logon uses the **ICAO code of the ATS unit** and must match the flight plan data (aircraft identification and registration). Errors in the flight plan or the FMS cause logon failures.
- Log on in good time before entering the CPDLC airspace, as published in the AIP.
- Only one ATS unit is the **current data authority** at any time — a controlled flight is under the control of only one ATC unit at a time (ICAO Annex 11).

**KEY TAKEAWAYS**

- DLIC (logon) comes first; then ACM, ACL and AMC — the four mandated services in Europe.
- DCL and DSC add departure and downstream clearances.
- Logon details must match the flight plan; only one current data authority at a time.

**KNOWLEDGE CHECK**

**What does the AMC service do?**

A. Sends weather reports

B. Lets the controller check that no aircraft is blocking the voice frequency

C. Assigns squawk codes

D. Manages the logon

**Answer: B.** The ATC Microphone Check service detects a stuck microphone blocking the frequency.

Training material only. It does not replace the Aircraft Flight Manual, FCOM, QRH, MEL or the Domestic Airlines Operations Manual, which always take precedence.